ResumeLens: Formal Language-Based Resume Screening

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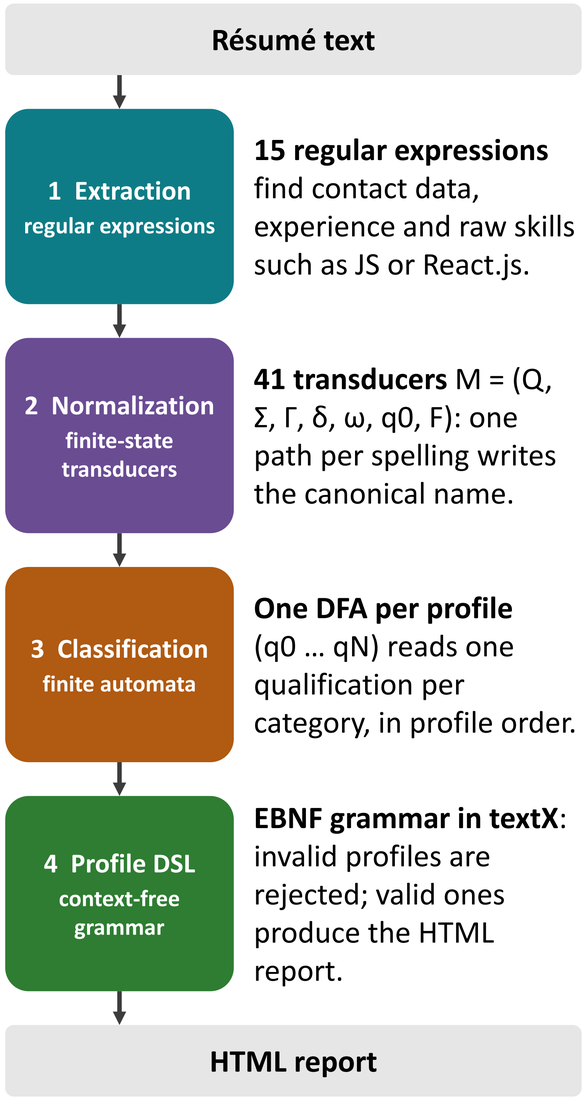
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Introduction

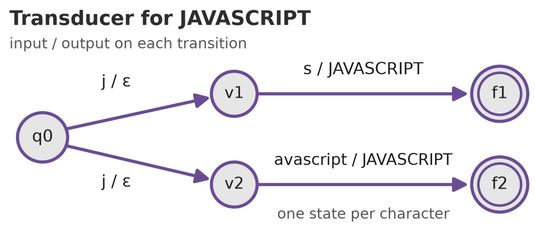
- The same skill is written many ways: **JS, Javascript, java script.**
- **Goal:** decide whether the qualifications in a résumé satisfy a formal pattern of a professional profile.
- It does not rank candidates or make hiring decisions.
- **Question:** can regular expressions, transducers, automata and a grammar solve each stage of the problem?
- **Four profiles, one engine:** Full Stack Developer, Machine Learning Engineer, DevOps Engineer and Data Engineer.
- Every result can be traced to a rule, a transducer path or an automaton transition.

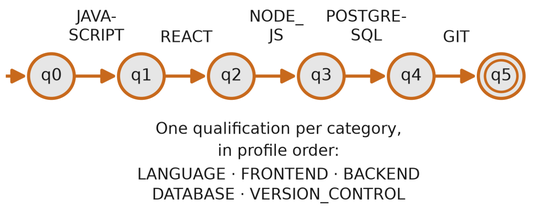
Methods

Each stage uses one formal model



Results





Assignment example

Git, NodeJS, JS, Postgres, React.js

→ **JAVASCRIPT, REACT, NODE_JS, POSTGRESQL, GIT**

Full Stack Developer: ACCEPTED

Sample résumés

- 7 résumés, 4 profiles: all classified as expected.
- One résumé accepted by two profiles, one by none.

Implementation

- **41 transducers, 4 DFAs, 115 passing tests.**

Discussion

Findings

- Profiles are data: one engine handles the four profiles.
- Sorting by profile category makes the result independent of résumé order.
- A DFA is enough: one qualification per category, in a fixed order.

Limitations

- Spellings are matched, not meanings, against a fixed catalog.
- Labeled fields and years in digits only.
- textX is PEG-based: close to, not identical to, a context-free grammar.

Future work

- ε-NFAs for optional categories and larger skill catalogs such as ESCO.

References

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